

Enterprise Windows Active Directory Infrastructure Lab

Designed and deployed a Windows Server 2022 enterprise infrastructure featuring Active Directory Domain Services (AD DS), DNS, DHCP, centralized authentication, and domain-joined client management.

Project Summary

The Enterprise Windows Active Directory Infrastructure Lab is a self-developed enterprise networking project built using Windows Server 2022 and Windows 11 virtual machines. The lab simulates a corporate network by deploying Active Directory Domain Services (AD DS), DNS, DHCP, centralized authentication, and domain-based client management. Windows clients automatically receive IP addresses through DHCP, resolve internal resources using DNS, and authenticate through Active Directory. The environment was validated through domain join verification, DHCP lease allocation, DNS name resolution, and network connectivity testing. This project demonstrates practical skills in Windows Server administration, enterprise networking, and Active Directory infrastructure deployment.

Technology Stack

Operating Systems

- Windows Server 2022
- Windows 11

Virtualization Platform

- Oracle VirtualBox

Server Roles

- Active Directory Domain Services (AD DS)
- DNS Server
- DHCP Server

Networking

- TCP/IP
- IPv4 Addressing
- DNS Name Resolution
- DHCP IP Address Allocation
- Domain Join

Administrative Tools

- Server Manager
- Active Directory Users and Computers
- DNS Manager
- DHCP Manager
- Command Prompt

Validation Commands

- ipconfig /all
- ping
- nslookup

Skills Demonstrated

- Windows Server Administration
- Active Directory Administration
- DNS Configuration
- DHCP Configuration
- Enterprise Networking
- Domain Administration

1. Active Directory Domain Services (AD DS)

Configured **Active Directory Domain Services (AD DS)** on Windows Server 2022 to establish a centralized domain environment. The server was promoted to a Domain Controller, enabling centralized authentication, user and computer management, and enterprise directory services.

Outcome

- Successfully deployed a functional Active Directory domain.
- Enabled centralized management of users and client systems.

2. DNS Server Configuration

Configured the DNS Server role to provide internal name resolution for the Active Directory environment. Created the Forward Lookup Zone and verified successful hostname resolution between domain resources.

Outcome

- Internal DNS name resolution operational.
- Successful hostname lookup using nslookup.

3. DHCP Server Configuration

Configured the DHCP Server role to automate IPv4 address allocation for client systems. Created and activated a DHCP scope and verified lease distribution to domain-joined clients.

Outcome

- Automatic IP address assignment.
- Successful DHCP lease allocation to CLIENT01.

4. Domain Join

Successfully joined the Windows 11 client (CLIENT01) to the Active Directory domain.

Outcome

- Client authenticated through Active Directory.
- Enterprise domain environment successfully established.

5. Enterprise Client Management

Configured the client workstation to operate as a managed enterprise device using centralized domain authentication and network services.

Outcome

- Centralized management.
- Domain-based authentication.
- Enterprise client configuration.

6. Network Configuration

Configured enterprise networking using IPv4 addressing, DNS configuration, DHCP services, and TCP/IP communication.

Outcome

- Stable communication between the Domain Controller and client.
- Reliable internal network operation.

7. Network Validation

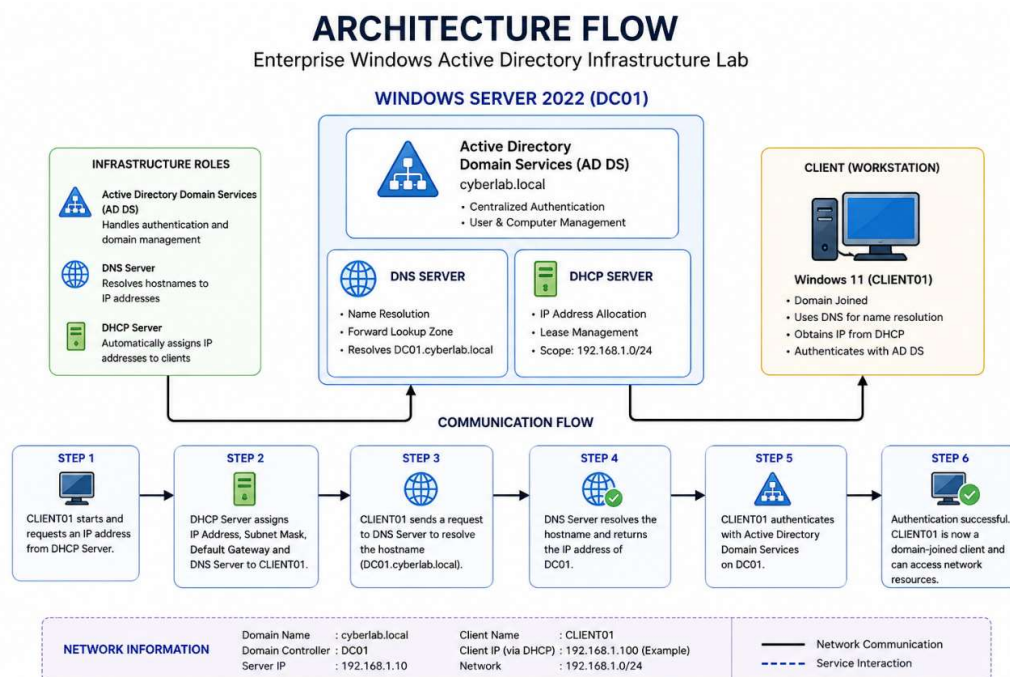
Performed validation using Windows networking utilities.

Validation included:

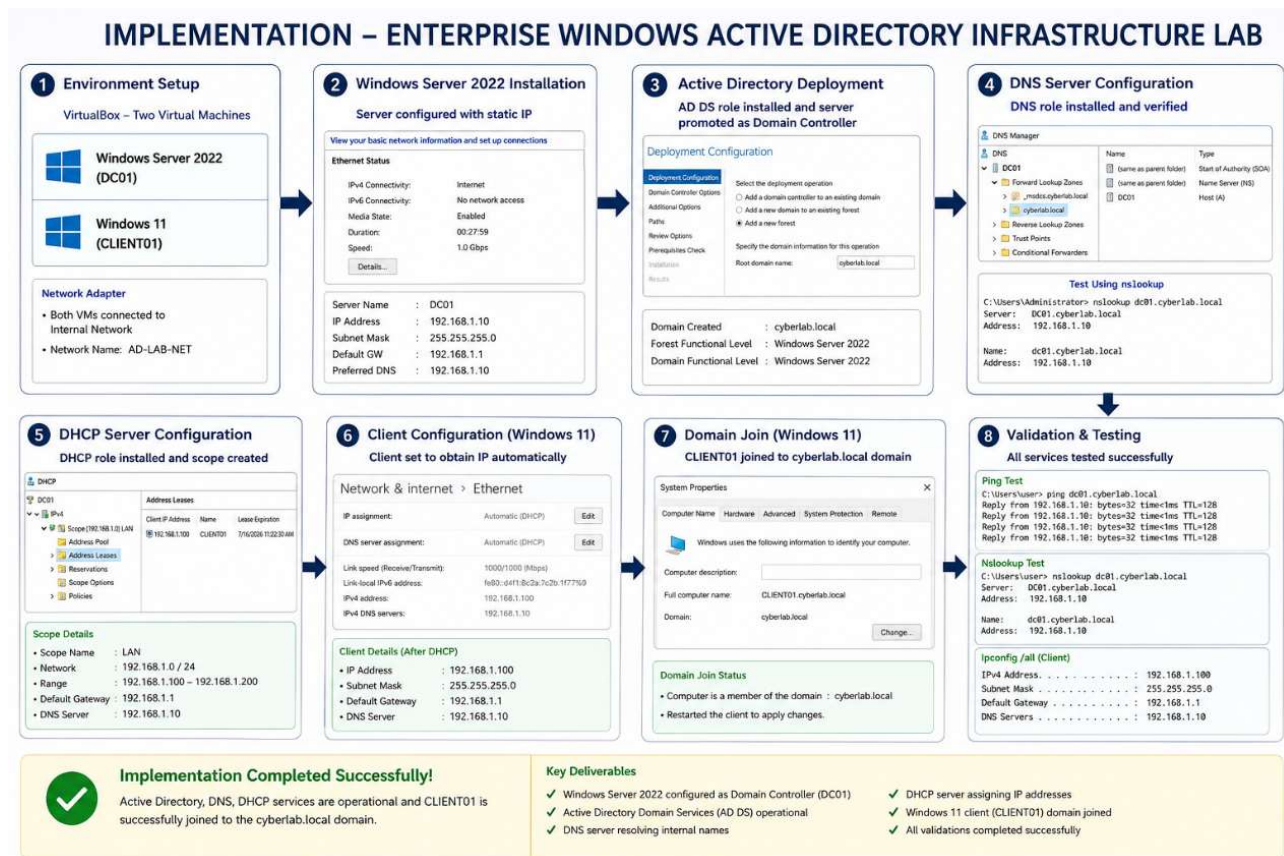
- ipconfig /all
- ping
- nslookup

Outcome

- Verified IP configuration.
- Verified DNS resolution.
- Verified client-to-server communication.
- Confirmed successful domain functionality.



Implementation Workflow



Challenges & Solutions

During the implementation of the Enterprise Windows Active Directory Infrastructure Lab, several technical challenges were encountered while deploying and integrating core Windows Server services. The primary challenges included configuring Active Directory Domain Services, establishing reliable DNS name resolution, setting up DHCP for automatic IP address allocation, and successfully joining the Windows 11 client to the Active Directory domain. Each issue was resolved through systematic troubleshooting using Windows Server management tools and built-in networking utilities such as ipconfig /all, ping, and nslookup. After validating each configuration step, all services operated successfully, resulting in a stable and fully functional enterprise infrastructure.

Lessons Learned

This project provided practical experience in deploying and managing a Windows-based enterprise network from the ground up. It reinforced the importance of the dependency between Active Directory, DNS, and DHCP services, and demonstrated how these components work together to provide centralized authentication, automated network configuration, and reliable name resolution. The project also highlighted the value of systematic validation and documentation throughout the deployment process. Overall, the implementation strengthened my understanding of Windows Server administration, enterprise networking, infrastructure troubleshooting, and the deployment of centralized IT services in a virtualized environment.